

ANSHAY AGARWAL

Senior AI Engineer | anshayagr@gmail.com | +91-8800472674 | [LinkedIn](#) | [GitHub](#)

SENIOR AI ENGINEER

Principal Vision & Imaging Engineer with 12+ years architecting ISP pipelines, computer vision, and AI-integrated perception systems. Ex-Qualcomm/Nvidia Lead Engineer who shipped imaging solutions on Snapdragon and Tegra platforms. Expert in ISP tuning (lens shading, Bayer demosaicing, HDR), system-level optimization (96% latency reduction), and ML model integration on heterogeneous compute architectures.

Stack: ISP Tuning & Imaging Pipelines, Computer Vision (OpenCV), Embedded SoC Optimization (Snapdragon/Tegra), C++, AI/ML Integration, Real-Time Signal Processing

EXPERIENCE

SCHOOLBOOK.AI | Lead AI Architect | AI, Python | Jun 2025 - Jun 2026

- **Engineered a real-time voice AI Teacher** capable of handling multiple student queries simultaneously with sub-second response times.
- **Architected a Multimodal RAG pipeline** utilizing Gemini Pro and a VectorDB to synthesize and teach directly from official NCERT educational materials
- **Engineered Autonomous Agentic Workflows:** Orchestrated multi-step reasoning chains using **LangGraph** and Python, enabling the AI to autonomously execute tool-use behaviors and dynamic planning.

QUALCOMM | Senior Lead Software Engineer | AI, Python, C++ | Apr 2023 - Mar 2025

- **Optimized XR Rendering Pipeline:** Reduced GPU Flicker correction latency by **96% (35ms → 1.3ms)**, directly enabling 90Hz refresh rates for Premium Tier Snapdragon chipsets.
- Spearheaded the design of novel HDR flows for Premium Tier chips, redefining image quality benchmarks for upcoming hardware.

GREYSCALEAI | Software Engineer | C++, Python, OpenCV | Feb 2021 - Mar 2023

- **Architected Multi-Modal Inspection Pipelines:** Synchronized high-throughput data streams from heterogeneous sensors (**X-Ray + RGB**), enabling real-time automated defect recognition.
- Designed a real-time embedded UI orchestrating communication between hardware sensors and user inputs.
- Led a complete codebase refactoring initiative that streamlined build processes.

NVIDIA | System Software Engineer | C++, Python | Aug 2017 - Jul 2018

- **Imaging Stack Optimization:** Enhanced the Tegra Camera imaging stack, resolving critical **Lens Shading** and **Bayer Demosaicing** artifacts for high-precision computer vision applications

DRDO | Scientist | C++, MATLAB, OpenCV | Aug 2013 - Jul 2015

- Engineered solutions for lens shading correction in Thermal Imagers (LREO/MREO), critical for night vision.
- Built a custom Raspberry Pi-based remote controller for Thermal Imagers, eliminating external procurement costs.

PROJECTS

EEG-to-Image Brain Signal Reconstruction | Python, CLIP

- Built neural decoding pipeline mapping EEG signals to visual imagery using CLIP-aligned mapping networks. Achieved 0.82 cosine similarity alignment between brain signal embeddings and image embeddings.

Vitalis | AI-Powered Medical Screening | Flutter, FastAPI, MedGemma, HeAR, AI-chat

- Built 8-module medical AI screening platform integrating Google's MedGemma, HeAR, and MedSigLIP foundation models on Cloud Run GPU backend.

Ai-Powered Youtube Video Summarizer | React Native, LangChain, Vector DB

- Developed a cross-platform mobile app (**Expo/React Native**) for podcast and video summarization using **LLMs** and **RAG**.
- Integrated vector databases for semantic search and managed full-stack deployment for real-time inference.

SKILLS

Languages: Python (10+), C++ (10+)

AI/ML: AI Agents, LangGraph, LLMs, RAG, VectorDB, Computer Vision (OpenCV), ASR (Whisper)

Systems: Git, Linux, FastAPI

ACHIEVEMENTS

- **Top Ranks:** GATE (AIR 423, 99.9%), IIT-JEE (Top 1%), GRE (321/340 - 170/170 Quant)

EDUCATION

IIT Delhi | M.Tech, Computer Technology | 2015 - 2017

- **Dissertation:** "Thermal Video Stabilization" for night vision systems.
- **Project:** Brain-Computer Interface (BCI) using EEG motor imagery signals for cursor control.

IIT Mandi | B.Tech, Electrical Engineering | 2009 - 2013

- **Dissertation:** Automobile Collision Prevention System
- **Leadership:** Founding Member of Robotics Section; Built IIT Mandi's first robot.

SPECIALIZATIONS

- **Wharton School:** Entrepreneurship Specialization (2021-2022)
- **DeepLearning.AI:** TensorFlow Professional Cert (2020), Deep Learning Specialization (2019)